TopoAML: transaction records become explicit account representations

Static, transductive account classification

A Observed transactions

Payer	Payee	Amount	Currency	Format
A	B	…	USD	Wire
A	C	…	USD	Wire
D	A	…	EUR	ACH

One row per transaction

TopoAML
account feature extractor

Account feature matrix

Account	T 16	G 16	M 5	K 3	E 4	P 12
A						
B						
…						

One row per account · Full = 56 columns

B Inside the account feature extractor

Fixed graph from all observed transactions; labels never enter extraction

Transaction layer T

16 columns

Counts / amounts
Counterparty diversity
Currencies / formats

Repeated / self transfers kept

Simple account graphs D and U

D

U: ignore direction

Directed structural S · 28

G 16
M 5 · K 3 · E 4

Directed reach / reconvergence / hubs
Untruncated local graph

Persistent homology P · 12

Radii 2 / 3 · sampled H0 / H1
Shared BFS prefix · cap 40

Feature blocks are schematic; S + P = 40 topology-aware descriptors. E is a deterministic transform of G.

Compare column subsets

Base: T (16)

Base+G: T + G (32)

Base+Struct: T + S (44)

Full: T + S + P (56)

Tabular classifier
LightGBM · XGBoost · random forest

Account risk ranking
Top-budget account alerts

Shared account splits; tune each input independently